Passage L20 Technical Notes

A scalable, bidirectional optical interconnect

Passage L20

The Passage™ L20 is the industry's first bidirectional-enabled high-density near-package and on-board optics solution, delivering 6.4 Tbps of optical bandwidth per direction for next-generation AI infrastructure.

Switch and XPU bandwidths are scaling faster than electrical signaling can follow. As AI clusters grow in scale and radix, the demand for high-bandwidth, low-latency optical interconnects has moved from the faceplate to the board itself. Pluggable optics cannot solve this because they contain bottlenecks at the panel, consume excessive power, and introduce retimer latency that degrades system performance.

The Passage L20 addresses this directly. Powered by Lightmatter's Passage photonic engine, the Passage L20 brings near-package optics (NPO) and on-board optics (OBO) to production-ready form factors, placing optical modules in close proximity to the host ASIC. The result is shorter electrical trace lengths, lower power consumption, and optical bandwidth that scales with deployment needs.

The Passage L20 is the first solution in its class to support bidirectional (BiDi) optical links to transmit and receive on a single fiber per lane. This halves fiber count requirements while maintaining full 6.4 Tbps per-direction bandwidth, making it the highest-density interconnect available for AI switching and accelerator fabrics.

The Interconnect Challenge

Conventional pluggable optics and electrical signaling cannot keep pace with the bandwidth demands of next-generation AI switches and accelerators.

AI infrastructure is scaling rapidly in both compute density and network radix. Each new generation of XPUs and switch ASICs demands significantly more I/O bandwidth, but the electrical and optical technologies that carry that bandwidth are struggling to keep up.

ELECTRICAL SIGNALING LIMITS

Higher radix designs require longer PCB traces between the ASIC and the panel. As trace lengths increase, signal integrity degrades, SerDes power consumption rises, and the practical bandwidth per lane decreases. Electrical signaling is approaching its physical limits for high-radix, high-bandwidth applications.

PLUGGABLE OPTICS LIMITS

Pluggable transceivers (OSFP, QSFP-DD) are constrained by the faceplate. Panel real estate is shared with power delivery and cooling infrastructure, placing a hard ceiling on total port count. Retimers required to bridge the electrical gap between the ASIC and the pluggable add latency and consume additional power. A single OSFP module occupies 88% more volume than an equivalent L20 module.

THE NEAR-PACKAGE OPPORTUNITY

Moving the optical engine off the faceplate and onto the board in close proximity to the ASIC eliminates the long trace penalty, removes the retimer from the critical path, and frees panel space for power and cooling. Near-package and on-board optics represent the next architectural step for AI networking infrastructure.

Near-Package Optics

The Passage L20 NPO configuration deploys directly on the host PCB adjacent to the ASIC, delivering high-bandwidth optical connectivity without modifying the chip package.

Near-Package Optics (NPO) places the Passage L20 module on the PCB or a mezzanine board in close proximity to the host ASIC. This shortens electrical trace lengths between the SerDes and the optical engine, reducing power consumption and improving signal integrity compared to pluggable solutions.

The NPO configuration requires no package modification to the host ASIC, making it accessible to a wide range of deployment scenarios without changes to the chip design or packaging process. The L20 module is reflow-compatible via its 1827-ball BGA form factor, integrating into standard PCB assembly workflows.

BIDI ADVANTAGE IN NPO DEPLOYMENTS

The Passage L20's bidirectional optical links transmit and receive on a single fiber per lane using Corning CPO Flexconnect. With 32 data fibers supporting 212.5 Gbps PAM4 per lane, the NPO configuration delivers 6.4 Tbps in each direction while halving the fiber infrastructure required compared to conventional duplex solutions. This density advantage is critical in high-radix switch deployments where fiber management is a practical constraint.

THERMAL MANAGEMENT

The L20 NPO module is cooled via cold plate, meeting ASHRAE A2 compliance requirements. This makes it compatible with standard data center cooling infrastructure without requiring specialized liquid cooling beyond what is already present in high-density AI rack deployments.

On-Board Optics

The Passage L20 OBO configuration uses a retimer to extend placement flexibility on the host PCB, enabling optimized board layout for thermal management and signal integrity in longer-reach applications.

On-Board Optics (OBO) addresses deployments where the Passage L20 module cannot be placed immediately adjacent to the host ASIC due to board layout constraints, thermal zones, or mechanical requirements. A retimer bridges the electrical path between the ASIC SerDes and the Passage L20, extending the allowable distance between the two without degrading signal integrity.

While the NPO configuration eliminates the retimer entirely, the OBO configuration trades that latency saving for significantly greater placement flexibility. This makes OBO the preferred choice for board designs where thermal management, power delivery, or mechanical constraints make direct ASIC proximity impractical.

BOARD LAYOUT BENEFITS

By decoupling the Passage L20 placement from the ASIC location, the OBO configuration allows system designers to optimize the board for airflow, power distribution, and mechanical constraints independently of optical module placement. This flexibility is particularly valuable in multi-ASIC designs where competing thermal and signal integrity requirements must be balanced across a complex PCB.

BANDWIDTH AND FIBER CONSISTENCY

The OBO configuration delivers identical optical bandwidth to the NPO configuration — 6.4 Tbps per direction across 32 data fibers at 212.5 Gbps PAM4 per lane. The BiDi single-fiber architecture and Corning CPO Flexconnect interface are consistent across both form factors, allowing system architects to select NPO or OBO based on board layout requirements without any change to the optical infrastructure or fiber plant.

Technical Specifications

Key electrical, optical, and mechanical specifications for the Passage L20 near-package and on-board optics module.

Parameter	Value
Optical bandwidth	6.4 Tbps per direction
SerDes line rate	212.5 Gbps PAM4
Data fibers	32 (Corning CPO Flexconnect)
Optical link type	Bidirectional (BiDi) — 1 fiber Tx/Rx
Form factor	1827-ball BGA, reflow compatible
Dimensions	37.5 mm x 26.4 mm
Power efficiency	5 pJ/bit
Volume vs OSFP	88% smaller by volume
Cooling	Cold plate, ASHRAE A2 compliant
Form factor options	Near-Package Optics (NPO), On-Board Optics (OBO)

ORDERING INFORMATION

The Passage L20 is available for design partnership engagements. Contact Lightmatter sales to discuss NPO and OBO configuration options, qualification timelines, and integration support.

Contact: lightmatter.co/contact-us